

# A Complex $M$ -Summand Obstruction for Symmetric Injective Tensors

Automatic math-discovery packet

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## Source Question

The source paper is Veronica Dimant, Daniel Galicer, and Ricardo Garcia, *Geometry of integral polynomials,  $M$ -ideals and unique norm preserving extensions*, arXiv:1108.3975. After proving that, over the real scalars, symmetric injective tensor powers never preserve nontrivial  $M$ -ideals, the authors ask the following.

Question 2.7. If  $X$  is a nontrivial  $M$ -ideal in  $Y$ ,  $X$  and  $Y$  complex Banach spaces, could  $\widehat{\bigotimes}_{\varepsilon_{k,s}}^{k,s} X$  be an  $M$ -ideal in  $\widehat{\bigotimes}_{\varepsilon_{k,s}}^{k,s} Y$ ?

ball is  $\{\pm\phi^k : \phi \in S_{X^*}\}$ , then it should be  $\mathcal{Q}({}^k X) = \mathcal{P}_I({}^k X)$ .

We have no idea how symmetric tensor products on  $M$ -ideals behave in the complex setting.

**Question 2.7.** *If  $X$  is a nontrivial  $M$ -ideal in  $Y$ ,  $X$  and  $Y$  complex Banach spaces, could  $\widehat{\bigotimes}_{\varepsilon_{k,s}}^{k,s} X$  be an  $M$ -ideal in  $\widehat{\bigotimes}_{\varepsilon_{k,s}}^{k,s} Y$ ?*

Figure 1: Question 2.7 in the source paper, page 10.

## Claim

We prove a partial negative answer: the phenomenon cannot occur for nontrivial complex  $M$ -summands. Thus any positive example to Question 2.7 would have to be a genuinely non-complemented  $M$ -ideal.

## Proof Intuition

The obstruction is already visible in  $\mathbb{C}e_1 \subset \ell_\infty^2$ . The symmetric injective tensor norm can be read as the supremum norm of  $k$ -homogeneous polynomials on the unit ball of  $(\ell_\infty^2)^* = \ell_1^2$ . The mixed monomial  $z^{k-1}w$ , with the right normalizing constant, has norm one. But adding any nonzero pure monomial  $\delta z^k$  makes the norm strictly larger than one, because at the point  $(|z|, |w|) = ((k-1)/k, 1/k)$  the mixed monomial is already extremal and the phase of  $w$  can align the two terms.

The three-ball property for  $M$ -ideals now fails: three pure monomials  $\omega_j z^k$ , with  $\omega_j$  the cubic roots of unity, cannot all be corrected by one pure monomial  $\lambda z^k$ . At least one difference  $\omega_j - \lambda$  has modulus at least one, forcing a uniform norm jump.

## Finite Obstruction

We use the standard three-ball characterization of  $M$ -ideals: if  $J$  is an  $M$ -ideal in  $E$ , then for every  $u \in B_E$ , every  $v_1, v_2, v_3 \in B_J$ , and every  $\eta > 0$ , there is  $v \in J$  such that

$$\|u + v_i - v\| \leq 1 + \eta, \quad i = 1, 2, 3.$$

**Lemma 1.** *Let  $k \geq 2$ , let  $E_k = \widehat{\bigotimes_{\varepsilon_{k,s}}^{k,s}} \ell_\infty^2$  over the complex scalars, and let*

$$J_k = \widehat{\bigotimes_{\varepsilon_{k,s}}^{k,s}} \mathbb{C}e_1 \subset E_k.$$

*Then  $J_k$  is not an  $M$ -ideal in  $E_k$ .*

*Proof.* Under the usual polynomial representation of the symmetric injective tensor norm,

$$\|p\|_{E_k} = \sup\{|p(z, w)| : |z| + |w| \leq 1\}.$$

The subspace  $J_k$  consists of the pure monomials  $\lambda z^k$ .

Set

$$C_k = \frac{k^k}{(k-1)^{k-1}}, \quad p_0(z, w) = C_k z^{k-1} w.$$

Since

$$\max_{0 \leq t \leq 1} t^{k-1}(1-t) = \left(\frac{k-1}{k}\right)^{k-1} \frac{1}{k},$$

we have  $\|p_0\| = 1$ .

Let

$$a_k = \left(\frac{k-1}{k}\right)^k.$$

For every  $\delta \in \mathbb{C}$ , choose  $|z| = (k-1)/k$  and  $|w| = 1/k$ , and choose the phase of  $w$  so that  $C_k z^{k-1} w$  and  $\delta z^k$  have the same argument. Then

$$\|p_0 + \delta z^k\| \geq 1 + a_k |\delta|.$$

Let  $\omega_1, \omega_2, \omega_3$  be the cubic roots of unity. For every  $\lambda \in \mathbb{C}$ ,

$$\max_{1 \leq j \leq 3} |\omega_j - \lambda| \geq 1,$$

because the smallest closed disk containing the three cubic roots of unity is the unit disk centered at the origin. Hence for every  $\lambda \in \mathbb{C}$ ,

$$\max_{1 \leq j \leq 3} \|p_0 + \omega_j z^k - \lambda z^k\| \geq 1 + a_k.$$

Taking  $u = p_0$  and  $v_j = \omega_j z^k \in B_{J_k}$ , the three-ball property fails, for instance with any  $0 < \eta < a_k$ . Therefore  $J_k$  is not an  $M$ -ideal in  $E_k$ .  $\square$

## M-Summand Consequence

**Theorem 1.** *Let  $k \geq 2$ , and let  $X$  and  $Z$  be nonzero complex Banach spaces. Put  $Y = X \oplus_\infty Z$ . Then the canonical subspace*

$$\widehat{\bigotimes_{\varepsilon_{k,s}}^{k,s}} X$$

*is not an  $M$ -ideal in*

$$\widehat{\bigotimes_{\varepsilon_{k,s}}^{k,s}} Y.$$

*Proof.* Choose  $x_0 \in S_X$ ,  $z_0 \in S_Z$ , and norming functionals  $x^* \in S_{X^*}$ ,  $z^* \in S_{Z^*}$  with  $x^*(x_0) = z^*(z_0) = 1$ . Define a norm-one projection

$$R : Y = X \oplus_\infty Z \longrightarrow \text{span}\{x_0, z_0\}$$

by

$$R(x, z) = x^*(x)x_0 + z^*(z)z_0.$$

The range is isometric to  $\ell_\infty^2$ , and  $R$  maps  $X$  onto  $\mathbb{C}x_0$ . Since the symmetric injective norm is functorial for contractions,  $\otimes^{k,s} R$  induces a contraction

$$\widehat{\bigotimes_{\varepsilon_{k,s}}^{k,s}} Y \longrightarrow \widehat{\bigotimes_{\varepsilon_{k,s}}^{k,s}} \text{span}\{x_0, z_0\}$$

which fixes the finite-dimensional tensor subspace generated by  $x_0, z_0$  and maps  $\widehat{\bigotimes_{\varepsilon_{k,s}}^{k,s}} X$  into the pure subspace generated by  $x_0^k$ .

Assume, toward a contradiction, that  $\widehat{\bigotimes_{\varepsilon_{k,s}}^{k,s}} X$  is an  $M$ -ideal in  $\widehat{\bigotimes_{\varepsilon_{k,s}}^{k,s}} Y$ . Apply the three-ball property to the finite obstruction from Lemma 1, transported to  $\text{span}\{x_0, z_0\}$ . Thus  $u$  is the normalized mixed monomial and  $v_1, v_2, v_3$  are the three pure root-of-unity monomials.

For  $0 < \eta < a_k$ , the three-ball property would give  $v \in \widehat{\bigotimes_{\varepsilon_{k,s}}^{k,s}} X$  such that

$$\|u + v_i - v\| \leq 1 + \eta, \quad i = 1, 2, 3.$$

Applying the contraction  $\otimes^{k,s} R$  gives a pure correction in the two-dimensional model with the same three inequalities, contradicting Lemma 1. Therefore the global  $M$ -ideal cannot exist.  $\square$

## Verification Notes

The proof uses only the standard polynomial formula for the symmetric injective norm on  $\ell_\infty^2$ , functoriality of the injective symmetric tensor norm for contractions, and the three-ball characterization of  $M$ -ideals. The accompanying script `code/verify_obstruction.py` checks the monomial normalization and the lower bound

$$\|p_0 + \delta z^k\| \geq 1 + \left(\frac{k-1}{k}\right)^k |\delta|$$

for several values of  $k$ .

## Limitations and Novelty Check

This packet is not a full solution to Question 2.7. It rules out all nontrivial complex  $M$ -summands, including the basic example  $\mathbb{C} \subset \ell_\infty^2$ . It does not rule out proper non-complemented complex  $M$ -ideals.

The run indexes were searched for arXiv:1108.3975, integral polynomials, complex  $M$ -ideals, symmetric injective tensor products, and unique norm-preserving extensions; no exact prior packet was found. A bounded web search for the exact question, the paper title, and later complex  $M$ -ideal/symmetric tensor keywords did not locate a later answer. The source paper itself proves the real case but explicitly leaves the complex case open.

## References

- V. Dimant, D. Galicer, and R. Garcia, *Geometry of integral polynomials,  $M$ -ideals and unique norm preserving extensions*, arXiv:1108.3975.
- P. Harmand, D. Werner, and W. Werner,  *$M$ -ideals in Banach spaces and Banach algebras*, Lecture Notes in Mathematics 1547, Springer, 1993. Used for the standard three-ball characterization of  $M$ -ideals.
- K. Floret, *Natural norms on symmetric tensor products of normed spaces*. Used as the standard reference for the symmetric injective tensor norm conventions followed by the source paper.

## Human Review Recommendation

Review the normalization convention for the symmetric tensor monomial representation against Floret/source notation. The argument is invariant under the usual symmetrization constants once the mixed monomial is normalized to norm one, but a human reviewer should check that the packet's polynomial model matches the source's  $\varepsilon_{k,s}$  convention exactly.